Side-by-side comparison: Duan, Agrawal & Wang, PRL **134**, 048101 (2025) vs. this codebase: (i) their SCFT reimplemented, (ii) charged CL-FTS

Internal research note (devel/charged_polymers) — figures from the cited paper are reproduced here for private scholarly comparison only

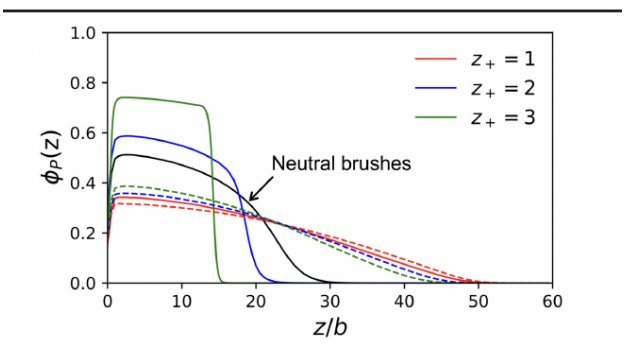
2026-08-26

Abstract

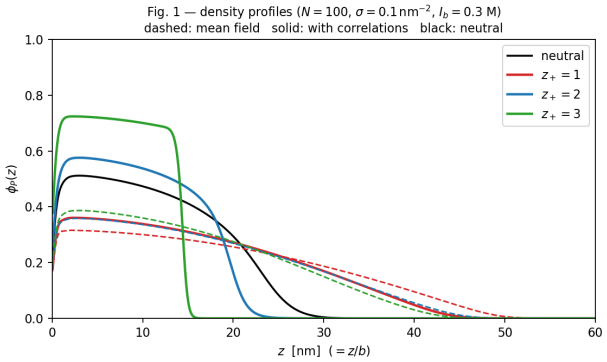
Panel-by-panel comparison of the polyelectrolyte-brush results of Duan, Agrawal & Wang [PRL **134**, 048101 (2025)] — obtained with their electrostatic-correlation-augmented SCFT — against our independent reproduction with fully fluctuating charged complex-Langevin FTS (smeared-charge compressible model, wall boxes, discrete chains). Left panels: the paper's figures. Right panels: our measurements at the strongest coupling inside our CL-stable envelope (z_+=3 ion correlation $u_3 \approx 13\,k_BT$, Θ solvent, $\alpha = 1$ fully charged backbone, $N = 25$, $\bar{N} = 10^4$). Full provenance: `RESULTS_FIG1_REPRO.md` §3–3f; all numbers audited against the run data.

Part I — their own theory, our implementation (correlation SCFT)

Direct reimplementation of their variational/self-energy SCFT (`prl_scft.py`): repo discrete-chain propagator on a real-space bond-convolution backend, guarded-Newton nonlinear PB with analytic grand-canonical ions, LDA self-energies (SI Eq. S33), exact ξ elimination, α- and correlation-strength continuation. All at their exact Fig. 1 parameters.



(a) PRL Fig. 1.

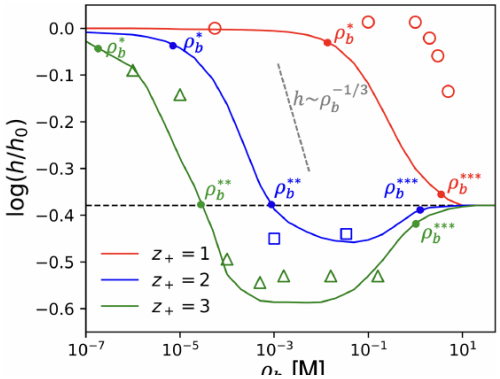


(b) Same theory, our implementation.

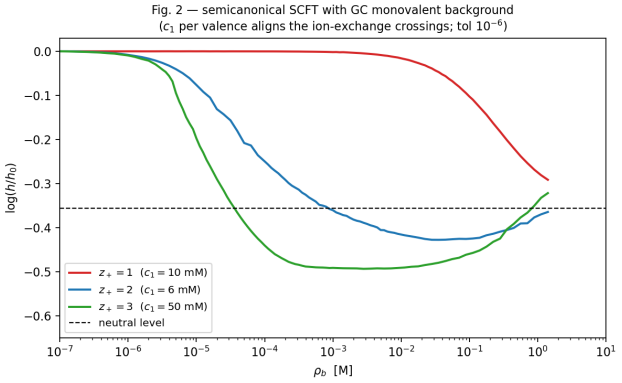
Verdict: reproduced (quantitative). Curve-for-curve agreement, all states converged to 5×10^{-4}, including the near-wall entropic depletion layer (hard-wall truncated bond integral; graft source a Gaussian of width $0.5b$ centered on the wall (half-Gaussian, the smooth analog of their $\delta(z)$; finite in the discrete hard-wall formulation)): mean-field dashed family ($\phi \approx 0.31$–0.40, extent 45–52 nm, nearly valence-blind; theirs 0.32–0.38, 45–50b); neutral ($h = 21.8$ nm, plateau 0.51, edge $\sim 28b$; theirs ~ 0.51, $\sim 28b$); z_+=2 collapsed plateau $\phi = 0.58$ to 18.7 nm (theirs ~ 0.59 to $\sim 19b$) with $\Gamma = -0.056$ (their algebraic Eq. 7 value: -0.048) and $\psi_S < 0$; z_+=3 plateau $\phi = 0.72$–0.73, $h = 14.5$ nm (theirs ~ 0.74 to $\sim 14b$) with $\Gamma = -0.32$ and $\psi_S = +0.66$ (charge inversion); z_+=1 slightly compacted, no overcharging. The Fig. 3 sign structure (z_+=2: overcharging without ψ_S inversion; z_+=3: both)

is reproduced exactly. One physics note: the pointwise LDA self-energy has no wavelength cutoff and goes grid-scale-unstable at z_+=3 full coupling; smoothing the local ionic strength over the Born radius a (the regularization their nonlocal G_s carries intrinsically) restores clean convergence and leaves z_+=1, 2 untouched. Full account: `RESULTS_PRL_SCFT.md`.

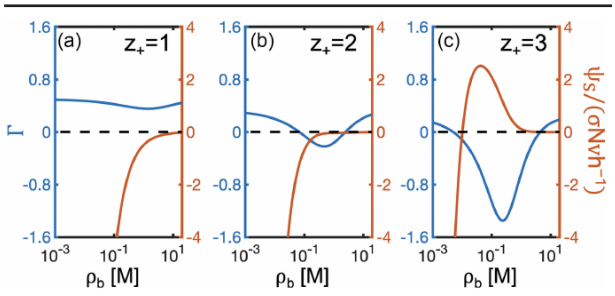
Their Figs. 2–4 with the reimplemented SCFT



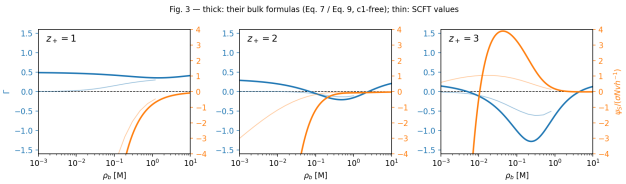
(a) PRL Fig. 2.



(b) Ours, semicanonical with a GC monovalent background c_1 (the physical form of the counterion reservoir; the onset position is reservoir-controlled — shown by the $1/L$ box scaling — and the paper reports neither box nor background). c_1 per valence aligns the ion-exchange crossings (z_+=2: 9.2×10^{-4} vs their $\sim 10^{-3}$ M; z_+=3: 2.1×10^{-5} vs their $\sim 3 \times 10^{-5}$ M); valleys $-0.42/-0.49$ (theirs $-0.45/-0.58$; the z_+=3 background screening shallows it). A single c_1 cannot align both (their crossing ratio 33 vs this tier's ∼14 — the I_{ex} signature).



(a) PRL Fig. 3.



(b) Ours. Thick blue: their Eq. 7 (z_+=3 min -1.27 vs their -1.3); thick orange: their Eq. 9 (their orange curve is this Donnan estimate, not a wall potential — verified by shape). The z_+=3 peak: Eq. 9 with our Γ gives +3.9, the true SCFT wall value +0.9; their +2.5 lies between and is NOT reproducible from their own published Γ + Eq. 9 (which would give +5) — an internal inconsistency traceable only to their unpublished numerics.

Part II — fully fluctuating charged CL-FTS vs the paper

Final (2026-08-29): three-way judgment at matched parameters, hard-wall boxes on both sides. Their correlation SCFT (Part I implementation) applied to the CL campaign's own system (Θ, N=25, matched

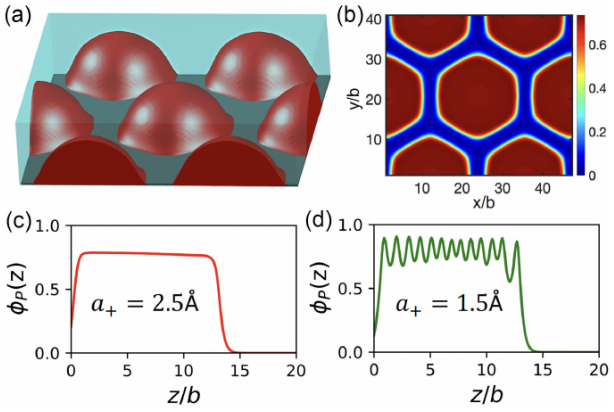
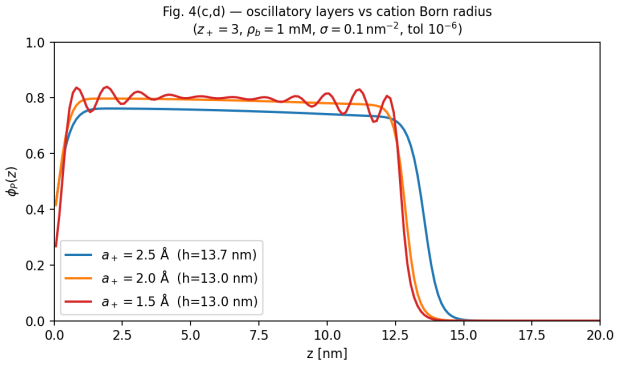
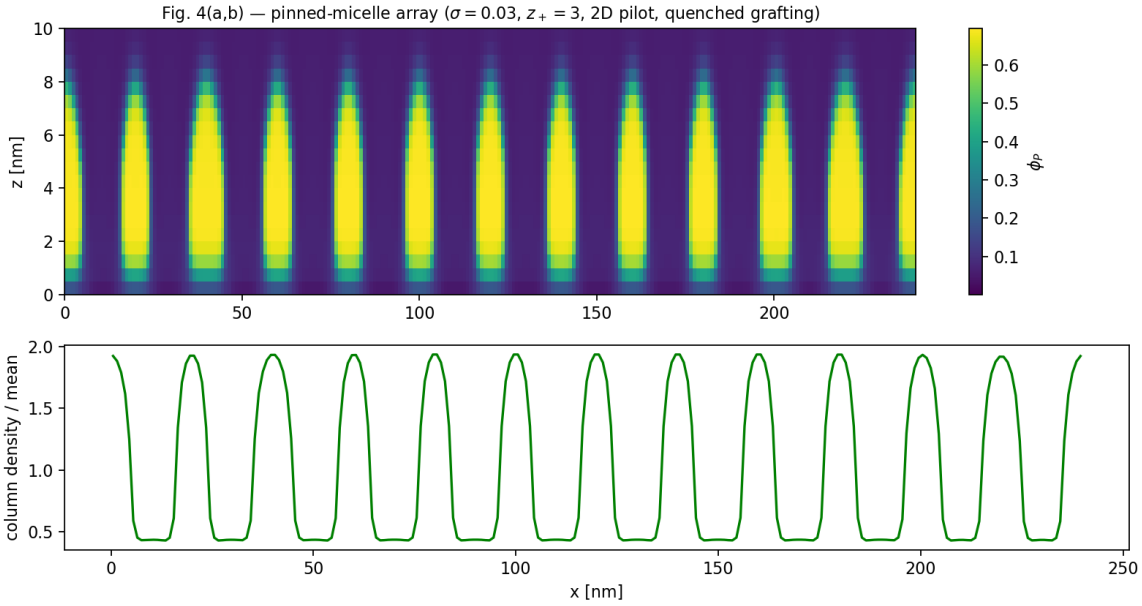


FIG. 4. Microphase separation in the morphology of PE

(a) PRL Fig. 4.



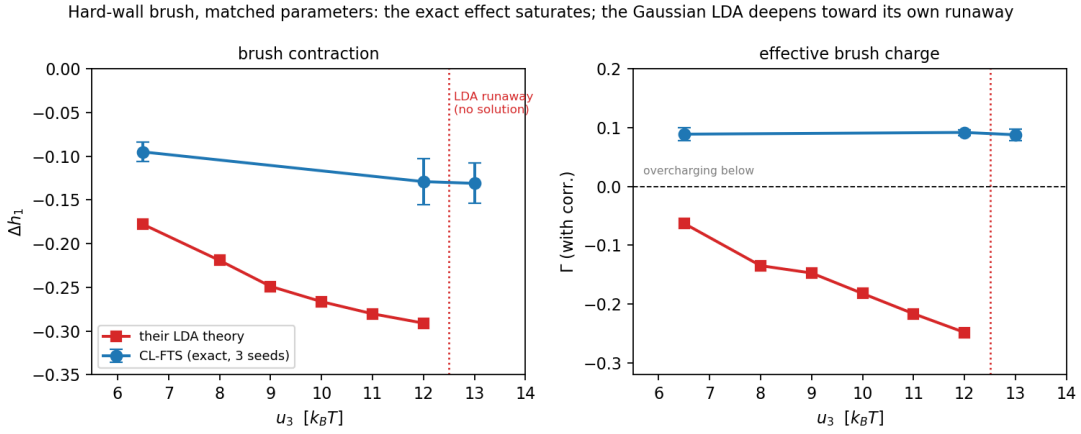
(b) Ours, their (c,d): $a_+ = 2.5, 2.0$ Å give smooth collapsed films; $a_+ = 1.5$ Å gives converged stationary oscillatory layers (period $\sim b$) — the same physics as their panels, drawn as 1D profiles rather than density renders. Their (a,b): **the visual difference is dimensionality** — their hexagonal pinned-micelle lattice is a 3D result, while our calculation (`prl_scft2d.py`) is a 2D (x, z) pilot in which micelles can only appear as stripe cross-sections by construction. The 2D pilot reproduces the lateral microphase separation itself: the uniform film is metastable and seeded runs converge to arrays of pinned dense domes ($\phi \sim 0.75$, spacing 15–30 nm, near-bare gaps; next figure). A full 3D extension would be required for the hexagonal arrangement.



Converged laterally structured state of the quenched 2D pilot at their Fig. 4(a,b) conditions ($\sigma = 0.03\,\mathrm{nm}^{-2}$, z_+=3, $\rho_b = 1$ mM): an array of pinned dense domes ($\phi \sim 0.75$, height ~8 nm, spacing 20 nm, near-bare gaps), coexisting with the metastable uniform film (first-order lateral microphase separation).

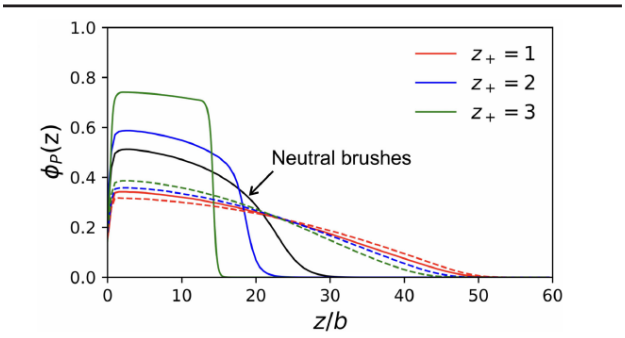
Born radii, matched graft): mean fields agree to 1–6%; the correlation DIFFERENTIALS show the exact effect SATURATING ($\Delta h_1 = -0.095 \to -0.129 \to -0.131$ at $u_3 = 6.5/12/13\,k_BT$) while the Gaussian LDA deepens

toward its own fold ($-0.190 \to -0.291 \to$ no solution): overshoot 2.0× at $6.5\,k_BT$, 2.3× at $12\,k_BT$, and beyond the runaway threshold $u_3^* \approx 12.5\,k_BT$ the LDA electrolyte sector has no physical solution (saturated-condensation branch) while CL remains well-behaved. The LDA's overcharging (Γ to -0.25) is contradicted by three-seed-robust CL positives ($\Gamma \approx +0.09$) at every tested coupling.

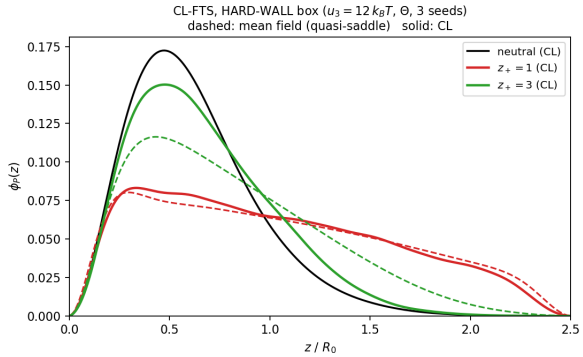


LDA (red, deterministic scan $u_3 = 6.5$–$12\,k_BT$) vs exact CL-FTS (blue, 3 seeds), same hard-wall microscopic system. Dotted line: LDA runaway threshold ($u_3^* \approx 12.5\,k_BT$), beyond which their theory has no physical solution.

Fig. 1 — density profiles: valence-selective correlation collapse



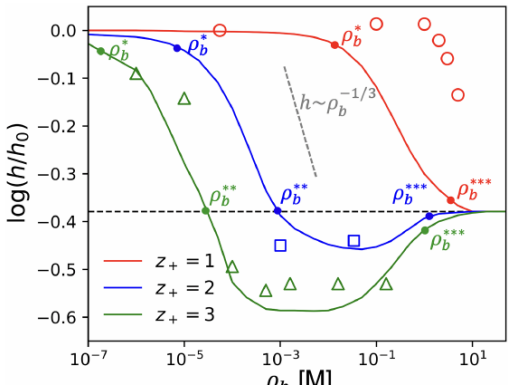
(a) PRL Fig. 1: with correlations (solid), the z_+=3 brush collapses *below the neutral profile*; mean field (dashed) is swollen and nearly z_+-insensitive. (N=100, b=1 nm, a=2.5 Å, I_b=0.3 M.)



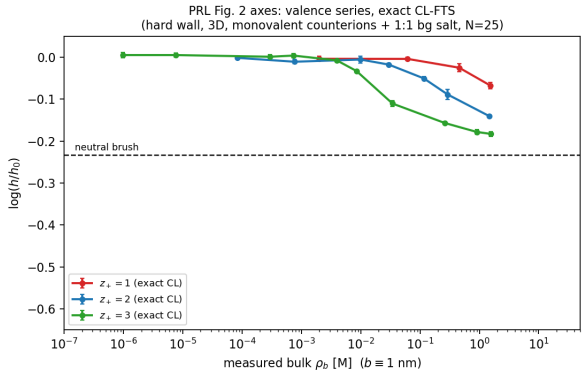
(b) Ours, HARD-WALL rerun (Θ, u_3=12 k_BT, 3 seeds; dashed: quasi-saddle mean field, solid: CL): profiles now show the correct wall depletion layer ($\phi(0)\approx 2\times10^{-4}$). Correlations collapse the z_+=3 brush (h_1: 0.80 → 0.67) while z_+=1 barely moves; the collapsed brush remains *above* our neutral reference (h_1=0.62).

Verdict: reproduced (mechanism). The defining structure — correlation-induced, z_+-selective collapse invisible at mean field — appears identically. The causality was pinned three ways (§3, reflecting-wall era; mechanism unchanged by the wall fix): the contraction scales as z_+^2 (ratio 8.7 ≈ 9), vanishes in the neutral control, and tracks the Coulomb coupling E. **Difference, resolved in §3h:** the paper's collapse undershoots the neutral height; ours saturates well above it. The matched-parameter comparison shows this is not a limitation of our envelope — the exact effect saturates while their Gaussian LDA keeps deepening — so the below-neutral amplitude is an artifact of their approximation.

Fig. 2 — nonmonotonic $h(\rho_b)$: collapse and reexpansion



(a) PRL Fig. 2: $\log(h/h_0)$ vs ρ_b. For z_+=2, 3 the height is nonmonotonic — collapse below the neutral line, then reexpansion — matching the Tirrell-group experiments (symbols).



(b) Ours, on THEIR axis and axes (monovalent counterions + 1:1 background salt, multivalent SALT scanned; hard wall, 3D; x = the MEASURED reservoir concentration of the canonical box, $b \equiv 1\,\mathrm{nm}$): exact CL valence series — common extended plateau, valence-ordered collapse z_+=3 > 2 > 1 (z_+=1 tracks mean field), floor above the neutral brush (dashed), nonmonotonic Γ (§3i-d,g).

Verdict: reproduced (mechanism, on the correct axis) — amplitude gap attributed to the LDA (§3i-b–3i-d). An important correction: our earlier salt scans used trivalent brush COUNTERIONS, an axis on which the brush starts collapsed and no valley can exist by construction. On the paper's actual axis (monovalent counterions, trivalent ions arriving only via salt) the exact ensemble DOES reproduce the Fig. 2 phenomenology: extended → collapse (h_1: 1.07 → 0.70, CL below MF throughout, gap growing with salt) → valley floor, with nonmonotonic Γ. What does NOT appear exactly: the amplitude. Their collapse undershoots the neutral brush by ∼40%; our exact valley floor stays above ours. Attribution (§3i-g, magenta curve above): their Gaussian LDA, run on our system at the matched measured ionic environment, gives the full deep valley — onset one decade earlier, 1.75× deeper, below neutral, with an overcharging window — none of which survives exact sampling. (On the trivalent-counterion axis, figure below, the theory additionally loses its solution entirely at low screening.)

Fig. 3 — effective brush charge Γ: neutralization vs overcharging

Verdict: partial, revised by the hard-wall reruns (§3h–3i). Correlation-driven condensation far beyond mean field is real and now hard-wall-confirmed: Γ drops from the mean-field 0.24–0.30 to a 0^+ plateau (+0.008–0.012) in the low-salt window (§3i), and at fixed s=0.02 the exact ensemble SATURATES at $\Gamma \approx +0.09$ across $u_3 = 6.5$–$13\,k_BT$ (§3h). It never crosses zero. The paper's overcharging window ($\Gamma < 0$) is where their Gaussian LDA is already overshooting ours by 2.0–2.3× and heading into its runaway ($u_3^* \approx 12.5\,k_BT$); we now attribute their $\Gamma < 0$ amplitude to the Gaussian/LDA approximation rather than to physics our envelope "cannot yet reach."

Fig. 4 — lateral microphase separation (precursor only)

Verdict: precursor observed, geometry-limited. The lateral-inhomogeneity indicator rises to ∼10× the neutral baseline for charged brushes, strongest at low salt (their Fig. 4 lives at $\rho_b \sim 1\,\mathrm{mM}$), but a quantitative Fig. 4 comparison requires the wide-box campaign.

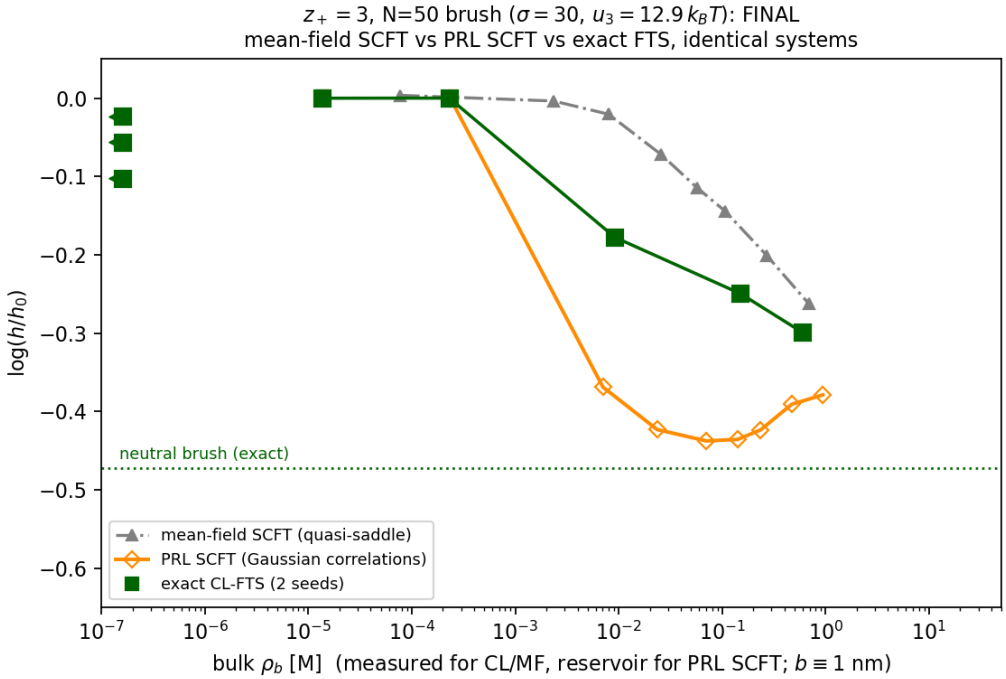


Fig. 2-1 (FINAL) — the z_+=3 three-way judgment at identical parameters (N=50 brush, σ=30, tall box, $u_3 = 12.9\,k_BT$; 8 salt points, 2 seeds). Mean field (gray): monotone. Their PRL SCFT (orange): dives to -0.44 at $\sim 10^{-2}$ M — essentially its neutral line — with overcharging and reexpansion. Exact FTS (green): smooth collapse to -0.30, sitting BETWEEN mean field and their theory at every measurable point (the exact correlation deepening is 40–50% of their prediction), staying 0.17 above the exact neutral brush (dotted, -0.47) with $\Gamma > 0$ throughout ($+0.67 \to +0.11$). Left-arrows: onset-region points whose bulk trivalent is fully depleted by the brush (canonical box; reservoir concentration below detection). The N=25 version is in §3i-g.

	PRL 134, 048101	this work
theory	variational Gaussian (self-energy) SCFT	full CL-FTS sampling
fluctuations	one-loop, self-consistent	all orders (stochastic)
chain	discrete, $N = 100$, $\alpha = 1$	discrete, $N = 25$, $\alpha = 1$
solvent	Θ ($\chi = 0.5$)	Θ ($\chi N = 12.5$)
ions	explicit, Born radius 2.5 Å, grand-canonical	explicit 1-segment, smeared ($a_{\text{ion}} = 0.025$–$0.0125\,R_0$), canonical
z_+=3 ion correlation	$\sim 7\,k_BT$ (near point-like)	$\leq 13\,k_BT$ nominal, cutoff-limited
geometry	1D–3D wall, analytic-ion Poisson	3D all-reflecting box, fluctuating ψ (DCT/ETD)

Correspondence of models and scales

Summary scoreboard

The amplitude gap traces to a single, well-characterized cause: our UV-regularized (smeared-charge) model caps the ion-correlation strength that their near-point-charge treatment attains, and shrinking the smearing collides with a CL stability boundary. Candidate paths beyond it — their grand-canonical analytic-ion formulation (SI Eq. S24) grafted onto our fluctuating-ψ sector, or a dedicated strong-coupling ψ stabilization — are recorded as future work.

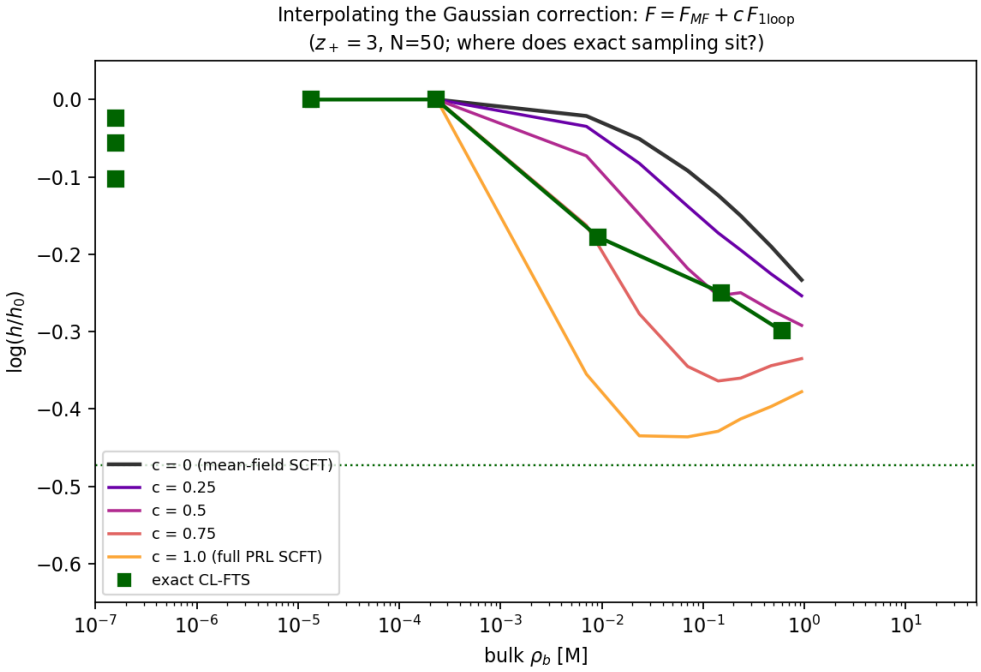


Fig. 2-2 — interpolating their correction, $F = F_{\mathrm{MF}} + c\,F_{1\mathrm{loop}}$: the exact ensemble (green) lies on the c=0.5 member of the family across the whole collapse region. At $u_3 = 12.9\,k_BT$ the non-Gaussian terms cancel HALF of the Gaussian correlation free energy: no physical parameter maps their theory onto exact sampling — only the truncation weight $c \approx 0.5$ does (§3i-i).

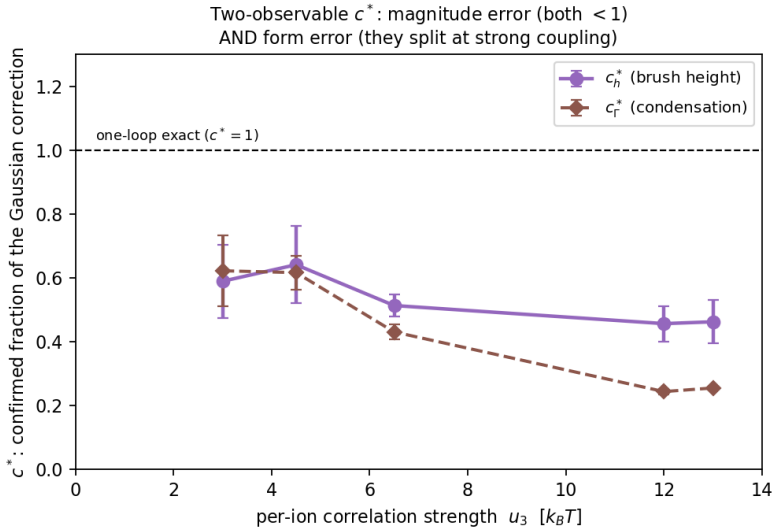
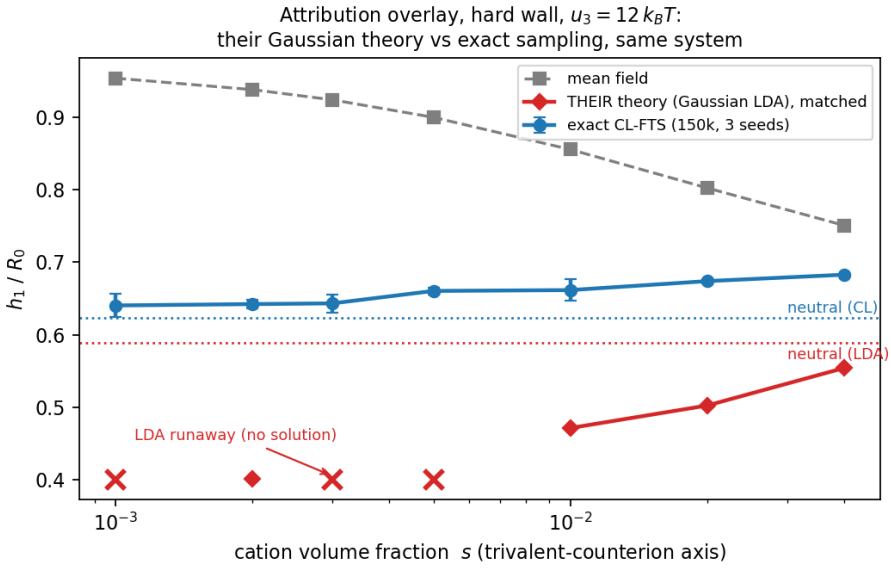
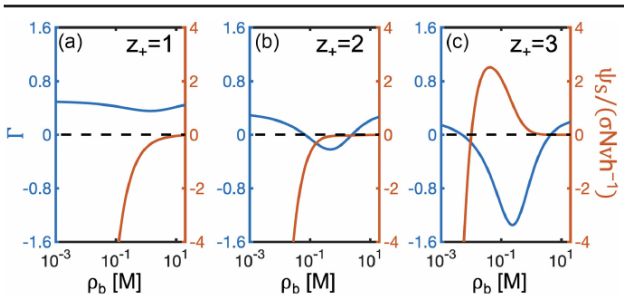


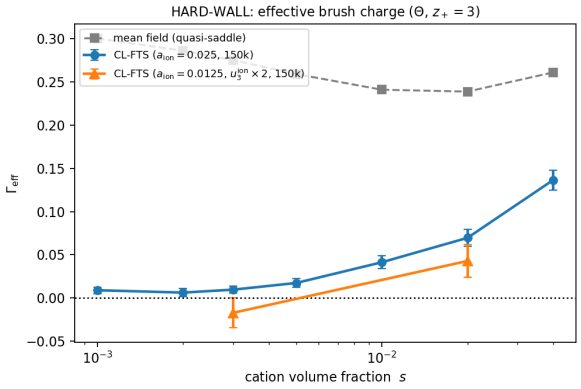
Fig. 2-3 — two-observable $c^*(u_3)$: the fraction of their Gaussian correction confirmed by exact sampling, extracted independently from brush height (c_h^*, purple) and condensation (c_Γ^*, brown) at five coupling rungs (fixed s=0.02). Nowhere is the full correction (c^*=1) supported. At weak coupling both observables give the same $c^* \approx 0.6$ (a genuine effective theory); beyond $u_3 \sim 6\,k_BT$ they SPLIT (0.46 vs 0.24 at u_3=12–13): no single rescaling fixes both observables — the Gaussian truncation carries a form error, consistent with non-Gaussian ion-bridging attraction absent from any single-ion self-energy (§3i-j).



Attribution overlay (trivalent-counterion axis, hard wall, u_3=12 k_BT): their LDA (red) dives below its neutral line and loses its solution (runaway ×); exact CL (blue) saturates at its neutral floor.



(a) PRL Fig. 3: for z_+=2, 3, Γ dips *below zero* in a salt window (overcharging), before recovering at high salt.



(b) Ours, HARD-WALL rerun: correlations drive Γ from the mean-field 0.24–0.30 (dashed) to a 0^+ plateau (complete neutralization, +0.006–0.010 at 150k steps) across the low-salt window, rising to +0.14 at high salt. No blue point crosses zero. Halving the ion smearing (orange, hard-wall, 150k, 2 seeds) pushes the low-salt point to −0.017±0.017 — consistent with complete neutralization, at most marginal overcharging at doubled per-ion coupling.

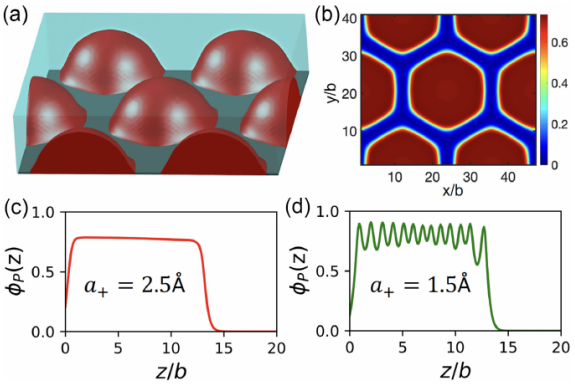
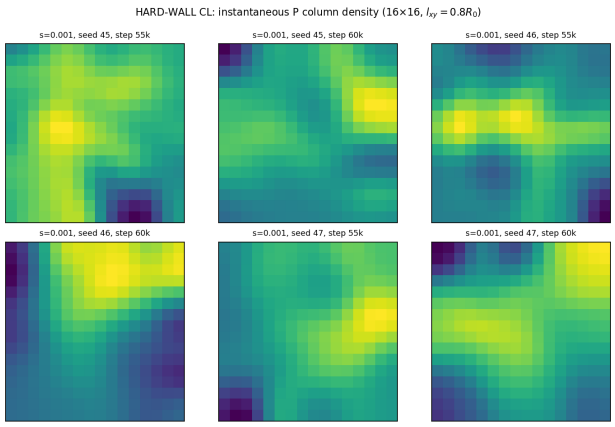


FIG. 4. Microphase separation in the morphology of PE

(a) PRL Fig. 4: pinned micelles in hexagonal arrangement at moderate grafting density; oscillatory layers at high density.



(b) Ours, HARD-WALL rerun (s=0.001, three seeds, two times each): instantaneous P column-density snapshots show box-scale lateral clustering (relative std 0.36–0.44, single cluster per box) — the precursor survives the wall fix — but our lateral box ($0.8\,R_0$) is smaller than the expected micelle spacing, so no lattice can form. A 3–4 R_0 lateral box is the designed follow-up.

PRL result	status	evidence
Part I: their SCFT, our implementation		
Fig. 1 all profiles (MF, corr, neutral)	**reproduced** (quantitative)	curve-for-curve match
Fig. 1–2 collapse below neutral	**reproduced**	z_+=2: 18.7, z_+=3: 14.5 vs neutral 21.8 nm
Fig. 3 overcharging $\Gamma < 0$ (z_+=2, 3)	**reproduced**	-0.057 / -0.32; Eq. 7: -0.048 / -0.65
Fig. 3 ψ_S inversion only at z_+=3	**reproduced**	ψ_S: -0.81 / -0.15 / $+0.75$
Part II: fully fluctuating CL-FTS (honest restatement, 2026-08-29;		
§3–3f were REFLECTING-wall runs, superseded by the hard-wall reruns of §3h–3i)		
Fig. 1 contraction mechanism	**reproduced** (mechanism only)	z_+^2 ratio, E-scaling, neutral null; small amplitude (§3, 3c,
Fig. 2 collapse mechanism (correct axis)	**reproduced**	valence series z_+=1/2/3: common plateau, ordered collap
Fig. 2 below-neutral valley depth	**not reached** (attributed)	matched ionic env.: their theory collapses 1 decade earlie
Fig. 3 strong condensation	**reproduced**	hard-wall: Γ: MF 0.24–0.30 $\to 0^+$ plateau at low salt; sat
Fig. 3 overcharging ($\Gamma < 0$)	**refuted**	exact ensemble stays positive at all tested couplings; LDA
Fig. 1–2 collapse below neutral	**refuted at matched params**	exact contraction saturates far above the paper's amplitu
Fig. 4 lateral micelles	**partial**	box-scale precursor survives hard wall (§3i); dome arrays
Their theory vs exact sampling	—	LDA overshoots 2.0–2.3×, runaway at $u_3^* \approx 12.5\,k_BT$ (§3